

Year 7 · Chapter 12 · Ratios, rates and time: answers (teacher copy)

12.01 Ratios (booklet p. 3)

Examples (model these live)

E1 A bag has 4 red and 7 blue counters. Write the ratio of red to blue.
red first · 4 : 7

E2 Write the ratio of shaded to unshaded squares.
3 shaded, 5 unshaded · 3 : 5

E3 In a ratio of 3 : 5, what fraction is the first part?
 $3 + 5 = 8$ parts · $\frac{3}{8}$

Questions (35 on today's skill, 12 from earlier lessons)

- | | | | |
|--------------------------|---|---|--|
| 1 12 : 9 | 15 $\frac{3}{4}$ | 28 HCF of 12 and 72 is 12 · = $\frac{1}{6}$ 4.01 | 38 No. The order matters: 5 : 4 compares Noah's to Jack's. |
| 2 5 : 5 | 16 $1\frac{3}{4}$ 4.01 | 29 games played: 25 · 12 : 25 | 39 blue: $\frac{3}{8} \cdot \frac{3}{8} \times 40 = 15$ L |
| 3 $\frac{6}{11}$ | 17 4 : 3 : 2 | 30 8 : 3 : 6 | 40 $3\frac{5}{8}$ 4.01 |
| 4 11 : 10 | 18 games played: 26 · 11 : 26 | 31 $1 \times 4 + 5 = 9$ · no: $9 \neq 11$ 9.06 | 41 No. The order matters: 9 : 8 compares Zac's to Aisha's. |
| 5 2 : 7 | 19 $25 \div 3 = 8$ r 1 · = $8\frac{1}{3}$ 4.01 | 32 blue: $\frac{3}{8} \cdot \frac{3}{8} \times 48 = 18$ L | 42 blue: $\frac{4}{6} \cdot \frac{4}{6} \times 24 = 16$ L |
| 6 $\frac{8}{12}$ | 20 4 : 8 : 6 | 33 blue: $\frac{2}{6} \cdot \frac{2}{6} \times 30 = 10$ L | 43 8 10.04 |
| 7 2 : 5 | 21 games played: 22 · 12 : 22 | 34 No. The order matters: 5 : 9 compares Tom's to Zac's. | 44 games played: 23 · 15 : 23 |
| 8 2 : 5 | 22 A (12 and 8) 10.08 | 35 $79 \div 8 = 9$ r 7 · = $9\frac{7}{8}$ 4.01 | 45 $\frac{53}{8}$ 4.01 |
| 9 6 : 1 | 23 5 : 4 : 5 | 36 blue: $\frac{3}{7} \cdot \frac{3}{7} \times 35 = 15$ L | 46 No. The order matters: 7 : 8 compares Kai's to Mia's. |
| 10 $\frac{9}{17}$ | 24 games played: 15 · 7 : 15 | 37 9 10.07 | 47 blue: $\frac{5}{8} \cdot \frac{5}{8} \times 16 = 10$ L |
| 11 4 : 2 | 25 3 : 3 : 7 | | |
| 12 6 cm^2 8.12 | 26 $\frac{19}{5}$ 4.01 | | |
| 13 7 : 7 : 9 | 27 blue: $\frac{5}{7} \cdot \frac{5}{7} \times 42 = 30$ L | | |
| 14 7 : 1 | | | |

12.02 Equivalent and simplified ratios (booklet p. 5)

Examples (model these live)

E1 Simplify 12 : 18.
HCF is 6 · = 2 : 3

E2 Simplify 40 cm : 2 m.
2 m = 200 cm · 40 : 200 = 1 : 5

E3 Find the missing number: 3 : 4 = $\square$: 20.
 $4 \times 5 = 20$ · $\square = 3 \times 5 = 15$

Questions (40 on today's skill, 16 from earlier lessons)

- | | | | |
|-----------------------------|--|--|--|
| 1 1 : 2 | 17 no 3.11 | 31 games played: 17 · 4 : 17 12.01 | 44 $\times 10: 6 : 5 \cdot = 6 : 5$ |
| 2 $\square = 25$ | 18 1 : 7 | 32 HCF is 5 · = 2 : 5 : 3 | 45 no 3.11 |
| 3 e.g. 2 : 18 and 10 : 90 | 19 HCF is 8 · = 5 : 3 | 33 2 m = 200 cm · 25 : 200 = 1 : 8 | 46 No. Multiply both parts by the same number: e.g. 12 : 27. |
| 4 7 : 3 | 20 2 L = 2000 mL · 200 : 2000 = 1 : 10 | 34 $C \rightarrow A, 3 \rightarrow 2 \cdot A2$ 9.01 | 47 games played: 19 · 14 : 19 12.01 |
| 5 $\square = 63$ | 21 7 : 8 : 5 12.01 | 35 $\times 2: 6 : 5 \cdot = 6 : 5$ | 48 $\times 6: 2 : 3 \cdot = 2 : 3$ |
| 6 e.g. 2 : 10 and 6 : 30 | 22 $\times 2: 2 : 1 \cdot = 2 : 1$ | 36 6 3.11 | 49 HCF is 2 · = 2 : 3 : 5 |
| 7 8 : 1 | 23 HCF is 15 · = 7 : 6 | 37 HCF is 8 · = 9 : 5 | 50 $\frac{3}{13} \cdot \approx 23.1\%$ 11.02 |
| 8 $\square = 24$ | 24 {S, C, H, O, L} 11.01 | 38 HCF is 4 · = 4 : 2 : 3 | 51 $\times 2: 5 : 6 \cdot = 5 : 6$ |
| 9 7 : 3 12.01 | 25 2 m = 200 cm · 10 : 200 = 1 : 20 | 39 $\times 10: 2 : 5 \cdot = 2 : 5$ | 52 1, 2, 4 3.11 |
| 10 $\square = 48$ | 26 $\times 2: 3 : 4 \cdot = 3 : 4$ | 40 8 : 8 : 6 12.01 | 53 $\times 8: 2 : 4 \cdot = 1 : 2$ |
| 11 e.g. 14 : 16 and 70 : 80 | 27 3 kg = 3000 g · 250 : 3000 = 1 : 12 | 41 No. Multiply both parts by the same number: e.g. 20 : 36. | 54 HCF is 5 · = 3 : 5 : 2 |
| 12 6 : 7 | 28 $\times 2: 9 : 8 \cdot = 9 : 8$ | 42 (0, 2) 9.06 | 55 9 : 8 : 9 12.01 |
| 13 8 9.04 | 29 1, 2, 4 3.11 | 43 HCF is 3 · = 1 : 2 : 3 | 56 $\times 2: 3 : 9 \cdot = 1 : 3$ |
| 14 HCF is 4 · = 6 : 1 | 30 HCF is 12 · = 3 : 8 | | |
| 15 $\square = 42$ | | | |
| 16 e.g. 6 : 16 and 24 : 64 | | | |

12.03 Ratio problems (booklet p. 7)

Examples (model these live)

E1 Share \$60 in the ratio 2 : 3.
5 parts: one part is \$12 · \$24 and \$36

E2 Boys to girls is 4 : 5. There are 20 boys. How many girls?
one part: $20 \div 4 = 5$ · girls: $5 \times 5 = 25$

E3 Cordial and water are mixed 1 : 6. How much cordial for 2.1 L of drink?
7 parts: $2.1 \div 7 = 0.3$ · cordial: 0.3 L

Questions (35 on today's skill, 14 from earlier lessons)

- | | | | |
|-----------------|----------------|-----------------------|---|
| 1 \$50 and \$40 | 5 \$8 and \$4 | 9 $\square = 9$ 12.02 | 12 4 parts: one part = $14 \cdot 28$ and 28 |
| 2 18 | 6 20 | 10 6 | 13 C 9.02 |
| 3 \$12 and \$6 | 7 \$27 and \$9 | 11 \$8 and \$12 | 14 8 |
| 4 5 | 8 20 | | |

- 15 \$12 and \$12
- 16 6 parts: one part = $21 \div 21$ and 105
- 17 $5 : 5$ ^{12.01}
- 18 8 parts: one part = $13 \cdot 65$ and 39
- 19 one part: $500 \div 2 = 250$ mL · water: $4 \times 250 = 1000$ mL
- 20 $\times 2: 2 : 6 \cdot = 1 : 3$ ^{12.02}
- 21 one part: \$5 · Jack: \$20
- 22 4 parts: one part = $14 \cdot 14$ and 42
- 23 $\frac{1}{3}$ ^{11.02}
- 24 one part: $600 \div 4 = 150$ mL · water: $7 \times 150 = 1050$ mL
- 25 one part: $900 \div 3 = 300$ mL · water: $9 \times 300 = 2700$ mL
- 26 one part: \$3 · Kai: \$12
- 27 $\frac{6}{11}$ ^{12.01}
- 28 8 parts: one part = $21 \cdot 147$ and 21
- 29 HCF is $9 \cdot = 2 : 9$ ^{12.02}
- 30 6 parts: one part = \$3 · \$3, \$6, \$9
- 31 one part: $500 \div 5 = 100$ mL · water: $6 \times 100 = 600$ mL
- 32 7 ^{10.04}
- 33 8 parts: one part = \$7 · \$7, \$21, \$28
- 34 difference: 1 part = 5, one part = 5 · total: $9 \times 5 = 45$
- 35 2 m = 200 cm · $50 : 200 = 1 : 4$ ^{12.02}
- 36 one part: $2000 \div 4 = 500$ g · flour: $7 \times 500 = 3500$ g
- 37 13 ^{9.04}
- 38 7 parts: one part = \$4 · \$8, \$16, \$4
- 39 difference: 3 parts = 15, one part = 5 · total: $7 \times 5 = 35$
- 40 $8 : 1$ ^{12.01}
- 41 one part: $1000 \div 2 = 500$ g · flour: $7 \times 500 = 3500$ g
- 42 difference: 3 parts = 12, one part = 4 · total: $7 \times 4 = 28$
- 43 one part: $1200 \div 3 = 400$ g · flour: $6 \times 400 = 2400$ g
- 44 no: 3 has a bigger sector ^{11.01}
- 45 8 parts: one part = \$7 · \$28, \$21, \$7
- 46 $\frac{4}{7}$ ^{12.01}
- 47 one part: \$4 · Jack: \$16
- 48 difference: 1 part = 4, one part = 4 · total: $5 \times 4 = 20$
- 49 HCF is $15 \cdot = 2 : 1$ ^{12.02}

12.04 Rates (booklet p. 9)

Examples (model these live)

E1 A car travels 240 km in 3 hours. Find its speed.

$$240 \text{ km} \div 3 \text{ h} \cdot = 80 \text{ km/h}$$

E2 4 kg of apples cost \$12. Find the cost per kilogram.

$$\$12 \div 4 \text{ kg} \cdot = \$3/\text{kg}$$

E3 A tap fills 90 L in 6 minutes. Write the rate in L/min.

$$90 \div 6 \cdot = 15 \text{ L/min}$$

Questions (37 on today's skill, 17 from earlier lessons)

- 1 99 km/h
- 2 \$6/kg
- 3 9 L/min
- 4 60 km/h
- 5 \$9/kg
- 6 21 L/min
- 7 105 km/h
- 8 \$9/kg
- 9 $\square = 10$ ^{12.02}
- 10 \$3/kg
- 11 14 L/min
- 12 69 km/h
- 13 21 cm^2 ^{8.12}
- 14 $\$78 \div 6 \text{ h} \cdot = \$13/\text{h}$
- 15 \$8/kg
- 16 21 L/min
- 17 8.89 ^{7.07}
- 18 69 km/h
- 19 $\$65 \div 5 \text{ h} \cdot = \$13/\text{h}$
- 20 $252 \div 4 \cdot = 63$ beats/min
- 21 one part: $600 \div 2 = 300$ mL · water: $8 \times 300 = 2400$ mL ^{12.03}
- 22 $350 \text{ km} = 3.5 \times 100 \text{ km} \cdot 31.5 \div 3.5 = 9 \text{ L/100 km}$
- 23 $\$95 \div 5 \text{ h} \cdot = \$19/\text{h}$
- 24 Cricket ^{10.01}
- 25 $162 \div 2 \cdot = 81$ beats/min
- 26 $360 \text{ km} = 3.6 \times 100 \text{ km} \cdot 28.8 \div 3.6 = 8 \text{ L/100 km}$
- 27 $2 : 6$ ^{12.01}
- 28 $\$60 \div 4 \text{ h} \cdot = \$15/\text{h}$
- 29 $\times 2: 7 : 5 \cdot = 7 : 5$ ^{12.02}
- 30 $50 \text{ m/s} = 180000 \text{ m/h} \cdot = 180 \text{ km/h}$
- 31 $156 \div 2 \cdot = 78$ beats/min
- 32 $1 \text{ m}^3 = 100 \times 100 \times 100 = 1\,000\,000 \text{ cm}^3 \cdot = 8\,000\,000 \text{ cm}^3$ ^{8.10}
- 33 $430 \text{ km} = 4.3 \times 100 \text{ km} \cdot 12.9 \div 4.3 = 3 \text{ L/100 km}$
- 34 0.9 ^{7.07}
- 35 $60 \text{ m/s} = 216000 \text{ m/h} \cdot = 216 \text{ km/h}$
- 36 $108 \text{ km/h} = 108000 \text{ m in } 3600 \text{ s} \cdot = 30 \text{ m/s}$
- 37 one part: \$5 · Noah: \$15 ^{12.03}
- 38 1 kg = $2 \times 500 \text{ g} \cdot \$8 \times 2 = \$16.00/\text{kg}$
- 39 {S, U, M, E, R} ^{11.01}
- 40 $30 \text{ m/s} = 108000 \text{ m/h} \cdot = 108 \text{ km/h}$
- 41 $90 \text{ km/h} = 90000 \text{ m in } 3600 \text{ s} \cdot = 25 \text{ m/s}$
- 42 $\frac{6}{10}$ ^{12.01}
- 43 1 kg = $5 \times 200 \text{ g} \cdot \$9 \times 5 = \$45.00/\text{kg}$
- 44 HCF is $15 \cdot = 4 : 5$ ^{12.02}
- 45 $72 \text{ km/h} = 72000 \text{ m in } 3600 \text{ s} \cdot = 20 \text{ m/s}$
- 46 1 kg = $5 \times 200 \text{ g} \cdot \$7 \times 5 = \$35.00/\text{kg}$
- 47 test x = 1, 2 and 3 in each · $y = 3x + 5$ ^{9.05}
- 48 $40 \text{ m/s} = 144000 \text{ m/h} \cdot = 144 \text{ km/h}$
- 49 1.9 ^{7.07}
- 50 $370 \text{ km} = 3.7 \times 100 \text{ km} \cdot 18.5 \div 3.7 = 5 \text{ L/100 km}$
- 51 1 kg = $4 \times 250 \text{ g} \cdot \$7 \times 4 = \$28.00/\text{kg}$
- 52 4 parts: one part = $21 \cdot 42$ and 42 ^{12.03}
- 53 $20 \text{ m/s} = 72000 \text{ m/h} \cdot = 72 \text{ km/h}$
- 54 A (16 and 11) ^{10.08}

12.05 Best buys (booklet p. 11)

Examples (model these live)

E1 500 g for \$4 or 750 g for \$5.25. Which is the better buy?

$$\$0.80 \text{ per } 100 \text{ g and } \$0.70 \text{ per } 100 \text{ g} \cdot 750 \text{ g for } \$5.25$$

E2 Find the cost of 1 pen when 6 pens cost \$4.50.

$$\$4.50 \div 6 \cdot = \$0.75$$

E3 A 2 L milk is \$3.60 and a 3 L milk is \$5.10. Which is better value?

$$\$1.80/\text{L and } \$1.70/\text{L} \cdot \text{ the } 3 \text{ L}$$

Questions (35 on today's skill, 14 from earlier lessons)

- 1 \$0.40
- 2 \$3.00/kg and \$2.50/kg
- 3 \$0.27
- 4 \$4.00/kg and \$3.50/kg
- 5 \$0.88
- 6 \$5.00/kg and \$5.50/kg
- 7 \$0.24
- 8 \$6.00/kg and \$6.50/kg
- 9 \$7/kg ^{12.04}
- 10 \$5.00/kg and \$5.50/kg
- 11 \$0.45
- 12 \$0.79 per 100 g and \$0.72 per 100 g · 1000 g is better
- 13 A2 ^{9.01}
- 14 \$5.00/kg and \$5.50/kg
- 15 \$0.83
- 16 \$0.77 per 100 g and \$0.83 per 100 g · 400 g is better
- 17 \$2.45/L and \$2.22/L · the 4 L bottle
- 18 games played: $19 \cdot 4 : 19$ ^{12.01}
- 19 \$0.92 per 100 g and \$0.82 per 100 g · 1000 g is better
- 20 \$1.85/L and \$1.60/L · the 3 L bottle
- 21 one part: $1000 \div 4 = 250$ mL · water: $9 \times 250 = 2250$ mL ^{12.03}
- 22 \$1.59/L and \$1.54/L · the 3 L bottle
- 23 \$0.88 per 100 g and \$0.97 per 100 g · 300 g is better
- 24 10 ^{11.04}
- 25 single: $8 \times \$5 = \40 · deal: $2 \times \$19 = \38 ; save \$2
- 26 $\frac{9}{12}$ ^{12.01}
- 27 \$1.76/L and \$1.53/L · the 2 L bottle
- 28 \$0.69 per 100 g and \$0.62 per 100 g · 600 g is better
- 29 $\square = 14$ ^{12.02}
- 30 single: $8 \times \$3 = \24 · deal: $2 \times \$11 = \22 ; save \$2
- 31 $450 \text{ km} = 4.5 \times 100 \text{ km} \cdot 22.5 \div 4.5 = 5 \text{ L/100 km}$ ^{12.04}

32 single: $6 \times \$3 = \18 · deal:
 $2 \times \$8 = \16 ; save \$2

33 two 3-packs: $\$10.80$ · the
3-pack (cheaper per item)

34 $162 = 3 \times 6 \times h = 18h$ · $h = 9$
cm ^{8.11}

35 single: $6 \times \$2 = \12 · deal:
 $2 \times \$5 = \10 ; save \$2

36 44 ^{7.08}

37 two 5-packs: $\$6.00$ · the
10-pack (cheaper per item)

38 single: $8 \times \$4 = \32 · deal:
 $2 \times \$15 = \30 ; save \$2

39 games played: $25 \cdot 12 : 25$
^{12.01}

40 two 3-packs: $\$8.20$ · the
3-pack (cheaper per item)

41 two 3-packs: $\$7.60$ · the
3-pack (cheaper per item)

42 single: $6 \times \$4 = \24 · deal:
 $2 \times \$11 = \22 ; save \$2

43 You cannot tell what is
being measured or in what
units. ^{10.02}

44 $\$2.49/\text{L}$ and $\$2.41/\text{L}$ · the
3 L bottle

45 1 : 7 ^{12.01}

46 two 6-packs: $\$10.80$ · the
12-pack (cheaper per item)

47 single: $10 \times \$2 = \20 ·
deal: $2 \times \$9 = \18 ; save \$2

48 e.g. 8 : 10 and 24 : 30 ^{12.02}

49 $\$1.71/\text{L}$ and $\$1.45/\text{L}$ · the
3 L bottle

12.06 Rate problems (booklet p. 13)

Examples (model these live)

E1 A car travels at 80 km/h for 3
hours. How far does it go?

$80 \times 3 = 240$ km

E2 How long does 150 km take at 60
km/h?

$150 \div 60 = 2.5 = 2$ h 30 min

E3 Petrol is $\$1.90/\text{L}$. Find the cost of
40 L.

$1.90 \times 40 = \$76$

Questions (36 on today's skill, 15 from earlier lessons)

1 94 km

2 $\$58.32$

3 $\$80$

4 276 km

5 $\$96.25$

6 $\$72$

7 356 km

8 $\$59.92$

9 $\$8/\text{kg}$ ^{12.04}

10 $\$95.40$

11 $\$75$

12 195 km

13 6 ^{8.09}

14 $200 \div 50 = 4 = 4$ h

15 $\$113.87$

16 $\$90$

17 $\$2.00/\text{kg}$ and $\$1.50/\text{kg}$
^{12.05}

18 425 km

19 $100 \div 40 = 2.5 = 2$ h 30
min

20 $64 \div 16 = 4$ minutes

21 games played: $18 \cdot 10 : 18$
^{12.01}

22 15 min = $\frac{15}{60}$ h = 0.25 h ·
 $20 \times 0.25 = 5$ km

23 $20 \div 40 = 0.5 = 30$ min

24 one part: $600 \div 2 = 300$
mL · water: $4 \times 300 = 1200$ mL
^{12.03}

25 $138 \div 23 = 6$ minutes

26 $56 \div 14 = 4$ minutes

27 20 min = $\frac{20}{60}$ h = 0.33 h ·

$24 \times 0.33 = 8$ km

28 (0, 3) ^{9.06}

29 $300 \div 100 = 3 = 3$ h

30 $\square = 24$ ^{12.02}

31 $198 + 82 = 280$ km · $280 \div$
 $4 = 70$ km/h

32 $105 \div 15 = 7$ minutes

33 10 L/min ^{12.04}

34 20 min = $\frac{20}{60}$ h = 0.33 h ·
 $20 \times 0.33 = 6.67$ km

35 $\$0.97$ per 100 g and
 $\$0.88$ per 100 g · 600 g is
better ^{12.05}

36 $168 + 93 = 261$ km · $261 \div 4$
 $= 65.3$ km/h

37 petrol: $8 \times 3.5 = 28$ L · cost:
 $28 \times 1.99 = \$55.72$

38 $\frac{1}{13} \approx 7.7\%$ ^{11.02}

39 $201 + 88 = 289$ km · $289 \div 4$
 $= 72.3$ km/h

40 $\$5$ and $\$20$ ^{12.03}

41 petrol: $7 \times 4.4 = 30.8$ L ·
cost: $30.8 \times 2 = \$61.60$

42 $114 + 168 = 282$ km · $282 \div$
 $4 = 70.5$ km/h

43 5 : 7 : 7 ^{12.01}

44 petrol: $7 \times 5.5 = 38.5$ L ·
cost: $38.5 \times 1.95 = \$75.08$

45 $198 + 94 = 292$ km · $292 \div 4$
 $= 73$ km/h

46 125 m^3 ^{8.11}

47 30 min = $\frac{30}{60}$ h = 0.5 h · 16
 $\times 0.5 = 8$ km

48 e.g. 10 : 6 and 50 : 30 ^{12.02}

49 petrol: $6 \times 5.5 = 33$ L · cost:
 $33 \times 1.83 = \$60.39$

50 $122 + 99 = 221$ km · $221 \div 3$
 $= 73.7$ km/h

51 $\$4.00/\text{kg}$ and $\$4.50/\text{kg}$
^{12.05}

12.07 Travel graphs (booklet p. 15)

Examples (model these live)

E1 How far from home is the car after
2 hours?

read up from 2 h · 120 km

E2 When did the car stop? For how
long?

flat part: 2 h to 3 h · 1 hour

E3 Find the speed in the first hour.

70 km in 1 h · 70 km/h

Questions (17 on today's skill, 6 from earlier lessons)

1 50 km

2 from 3 h to 4 h

3 80 km

4 from 3 h to 4 h

5 90 km

6 $80 - 40 = 40$ km in 1 h ·
40 km/h

7 $130 - 80 = 50$ km in 1 h ·
50 km/h

8 steepest part: 0 h to 1 h ·
70 km in that hour

9 games played: $17 \cdot 8 : 17$
^{12.01}

10 steepest part: 0 h to 1 h ·
60 km in that hour

11 $150 - 80 = 70$ km in 1 h ·
70 km/h

12 49 ^{10.06}

13 160 km in 4 h · $160 \div 4 =$
40 km/h

14 $\square = 24$ ^{12.02}

15 150 km in 4 h · $150 \div 4 =$
37.5 km/h

16 read across from 70 km ·
between 1 h and 2 h

17 9 values: the 5th · median
47 ^{10.07}

18 160 km in 4 h · $160 \div 4 =$
40 km/h

19 read across from 60 km ·
between 1 h and 2 h

20 140 km in 4 h · $140 \div 4 =$
35 km/h

21 $\frac{5}{6}$ ^{11.05}

22 steepest part: 0 h to 1 h ·
60 km in that hour

23 $\frac{6}{14}$ ^{12.01}

12.08 Time (booklet p. 17)

Examples (model these live)

E1 Change 2.25 hours to hours and
minutes.

$0.25 \times 60 = 15 = 2$ h 15 min

E2 Add 2 h 45 min and 1 h 30 min.

3 h 75 min · = 4 h 15 min

E3 Change 200 minutes to hours and
minutes.

$200 \div 60 = 3$ r 20 · 3 h 20 min

Questions (36 on today's skill, 14 from earlier lessons)

1 300 min

2 2 h 33 min

3 1 h 6 min

4 360 min

5 4 h 2 min

6 2 h 15 min

7 60 min

8 2 h 39 min

9 50 ^{8.01}

10 3 h 13 min

11 3 h 45 min

12 240 min

- 13 Swim ^{10.01}
 14 3 h 48 min · = 3 h 48 min
 15 1 h 44 min
 16 3 h 6 min
 17 5 h 11 min · = 5 h 11 min
 18 change to minutes if needed · = 3 h 29 min
 19 games played: 18 · 14 : 18 ^{12.01}
 20 $28 \div 60 \approx 0.47 \approx 1.47$ h
 21 5 h 64 min · = 6 h 4 min
 22 $225 \div 25 = 9$ minutes ^{12.06}
 23 change to minutes if needed · = 1 h 32 min
 24 change to minutes if needed · = 3 h 59 min
 25 $2 \div 60 \approx 0.03 \approx 1.03$ h
 26 A ^{9.02}
 27 2 h 40 min · = 2 h 40 min
 28 $\square = 12$ ^{12.02}
 29 $714 \div 60 = 11$ r 54 · 11 min 54 s
 30 change to minutes if needed · = 2 h 58 min
 31 \$4.00/kg and \$3.50/kg ^{12.05}
 32 $1 \div 60 \approx 0.02 \approx 3.02$ h
 33 $2 \times 60 = 120$ · 120 + 33 = 153 min ^{8.01}
 34 $657 \div 60 = 10$ r 57 · 10 min 57 s
 35 $5 \times 43 = 215$ min · = 3 h 35 min
 36 9 values: the 5th · median ^{10.07}
 37 No. 0.5 h is half of 60 min = 30 min, so 4 h 30 min.
 38 80 km ^{12.07}
 39 $734 \div 60 = 12$ r 14 · 12 min 14 s
 40 $3 \times 44 = 132$ min · = 2 h 12 min
 41 $5 \times 34 = 170$ min · = 2 h 50 min
 42 $290 \div 60 = 4$ r 50 · 4 min 50 s
 43 39 ^{10.06}
 44 $32 \div 60 \approx 0.53 \approx 2.53$ h
 45 \$6/kg ^{12.04}
 46 $5 \times 47 = 235$ min · = 3 h 55 min
 47 $728 \div 60 = 12$ r 8 · 12 min 8 s
 48 \$0.24 ^{12.05}
 49 $49 \div 60 \approx 0.82 \approx 2.82$ h
 50 cm to mm: $\times 10 \cdot 16 \times 10 = 160$ mm ^{8.01}

12.09 12-hour and 24-hour time (booklet p. 19)

Examples (model these live)

- E1 Write 3:45 pm in 24-hour time.
 $3 + 12 = 15$ · 1545
 E2 Write 0820 in 12-hour time.
 before noon · 8:20 am
 E3 Write 2310 in 12-hour time.
 $23 - 12 = 11$ · 11:10 pm

Questions (36 on today's skill, 14 from earlier lessons)

- 1 0615
 2 8:40 pm
 3 1530
 4 6:35 pm
 5 0250
 6 8:35 am
 7 0230
 8 2:35 am
 9 3 h 13 min ^{12.08}
 10 5:00 am
 11 1605
 12 $14 - 12 = 2$ · 2:02 pm
 13 $\frac{5}{6}$ ^{11.05}
 14 6:10 pm
 15 1545
 16 $17 - 12 = 5$ · 5:48 pm
 17 from 3 h to 4 h ^{12.07}
 18 4:35 pm
 19 $15 - 12 = 3$ · 3:13 pm
 20 $1956 + 1$ h 4 min · 2100
 21 games played: 26 · 11 : 26 ^{12.01}
 22 $19 - 12 = 7$ · 7:12 pm
 23 $1929 + 1$ h 17 min · 2046
 24 $84 \div 21 = 4$ minutes ^{12.06}
 25 $23 - 12 = 11$ · 11:15 pm
 26 $2206 + 1$ h 8 min · 2314
 27 $20 - 12 = 8$ · 8:13 pm
 28 You cannot tell what is being measured or in what units. ^{10.02}
 29 7:07 am to 2:22 pm · 7 h 15 min
 30 $\square = 8$ ^{12.02}
 31 $1541 + 35$ min · 1616
 32 $16 - 12 = 4$ · 4:16 pm
 33 4 h 30 min ^{12.08}
 34 9:04 am to 2:02 pm · 4 h 58 min
 35 10:13 am to 7:17 pm · 9 h 4 min
 36 $2339 + 3$ h 31 min = 0310 (next day) · 3:10 am
 37 4 ^{10.04}
 38 10:18 am to 8:30 pm · 10 h 12 min
 39 \$144 ^{12.06}
 40 $2102 + 2$ h 23 min = 2325 (next day) · 11:25 pm
 41 7:37 am to 4:32 pm · 8 h 55 min
 42 one part: $200 \div 1 = 200$ mL · water: $4 \times 200 = 800$ mL ^{12.03}
 43 $2303 + 1$ h 57 min = 0100 (next day) · 1:00 am
 44 11:35 am to 6:24 pm · 6 h 49 min
 45 {S, U, M, E, R} ^{11.01}
 46 $1034 + 39$ min · 1113
 47 \$9/kg ^{12.04}
 48 $2121 + 2$ h 26 min = 2347 (next day) · 11:47 pm
 49 8:39 am to 3:05 pm · 6 h 26 min
 50 from 3 h to 4 h ^{12.07}

12.10 Time differences (booklet p. 21)

Examples (model these live)

- E1 How long is it from 9:40 am to 2:15 pm?
 $9:40 \rightarrow 10:00$: 20 min, $\rightarrow 2:00$: 4 h · + 15 min = 4 h 35 min
 E2 Someone was born in 2013. How old are they on their birthday in 2026?
 $2026 - 2013 = 13$
 E3 A film starts at 7:25 pm and runs 1 h 50 min. When does it end?
 $7:25 + 1$ h = 8:25 · + 50 min = 9:15 pm

Questions (36 on today's skill, 14 from earlier lessons)

- 1 2 h 30 min
 2 16
 3 4 h 15 min
 4 18
 5 4 h 30 min
 6 11
 7 1 h
 8 13
 9 1:00 am ^{12.09}
 10 15
 11 1 h 45 min
 12 count on to 9:00 am: 57 min · total: 3 h 19 min
 13 A (9 and 8) ^{10.08}
 14 17
 15 1 h 15 min
 16 count on to 12:00 pm: 10 min · total: 4 h 7 min
 17 3 h 12 min ^{12.08}
 18 14
 19 count on to 10:00 am: 34 min · total: 2 h 16 min
 20 5:55 pm + 1 h 41 min · 7:36 pm
 21 games played: 24 · 14 : 24 ^{12.01}
 22 count on to 9:00 am: 38 min · total: 4 h 7 min
 23 8:05 pm + 2 h 27 min · 10:32 pm
 24 8:55 pm + 2 h 3 min · 10:58 pm
 25 count on to 10:00 am: 23 min · total: 5 h 41 min
 26 likely ^{11.03}
 27 $2026 - 1977 = 49$ · 48 (birthday not yet)
 28 $\square = 24$ ^{12.02}
 29 6:55 pm + 1 h 41 min · 8:36 pm
 30 count on to 12:00 pm: 23 min · total: 4 h 15 min
 31 1930 ^{12.09}
 32 $2026 - 1983 = 43$ · 43 (birthday has passed)
 33 $18 \div 60 \approx 0.3 \approx 2.3$ h ^{12.08}
 34 $2026 - 1984 = 42$ · 42 (birthday has passed)
 35 to midnight: 30 min · + 5 h = 5 h 30 min
 36 C $\rightarrow$ A, 5 $\rightarrow$ 3 · A3 ^{9.01}
 37 $2026 - 1985 = 41$ · 40 (birthday not yet)
 38 60 km ^{12.07}
 39 to midnight: 2 h 45 min · + 6 h = 8 h 45 min
 40 $2026 - 1957 = 69$ · 68 (birthday not yet)
 41 one part: $300 \div 1 = 300$ mL · water: $9 \times 300 = 2700$ mL ^{12.03}
 42 to midnight: 1 h 15 min · + 5 h = 6 h 15 min
 43 $2026 - 1966 = 60$ · 59 (birthday not yet)
 44 $\frac{3}{4}$ ^{11.02}
 45 8:05 pm + 1 h 56 min · 10:01 pm

46 \$9/kg 12.04

47 to midnight: 30 min · + 7 h
30 min = 8 h

48 $2026 - 1986 = 40 \cdot 39$
(birthday not yet)

49 240 min 12.08

50 7:05 pm + 2 h 7 min · 9:12
pm

12.11 Timetables (booklet p. 23)

Examples (model these live)

E1 When does the first bus reach the
Beach?

first column, Beach row · 0736

E2 How long does the bus take from
the Station to the Mall?

0700 to 0745 · 45 minutes

E3 You get to the Library at 0730.

When is the next bus to the Mall, and
when does it arrive?

0757 from the Library · arrives 0830

Questions (20 on today's skill, 8 from earlier lessons)

1 0807

2 0844

3 0623

4 0903

5 0713

6 count on from one time
to the other · 34 minutes

7 0657

8 4.148 L 8.13

9 count on from one time
to the other · 40 minutes

10 count on from one time
to the other · 21 minutes

11 games played: 18 · 15 : 18
12.01

12 count on from one time
to the other · 37 minutes

13 count on from one time
to the other · 34 minutes

14 next bus at the Library:
0910 · reaches the Mall at
0939

15 {P, E, N, C, I, L} 11.01

16 count on from one time
to the other · 38 minutes

17 $\square = 5$ 12.02

18 next bus at the Library:
0751 · reaches the Mall at
0823

19 next bus at the Library:
0911 · reaches the Mall at
0946

20 arrivals at the Mall: 0801,
0846, 0931 · the 0800 bus

21 in order: 13, 33, 34, 37, 42,
46, 58 · median 37, range 45
10.06

22 next bus at the Library:
0907 · reaches the Mall at
0934

23 2 h 45 min 12.08

24 arrivals at the Mall: 0636,
0721, 0806 · the 0645 bus

25 next bus at the Library:
0659 · reaches the Mall at
0733

26 10 10.04

27 count on from one time
to the other · 30 minutes

28 \$7/kg 12.04